Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · FACTORIAL ANOVA

main effects + interaction of 2+ factors

Table 1. Cell descriptives

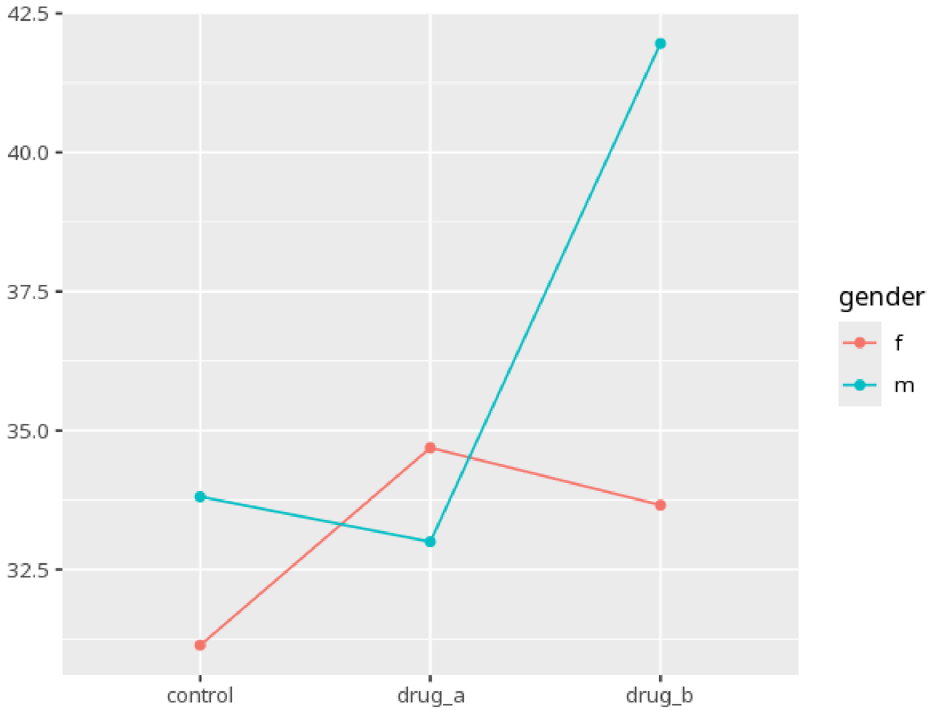
Factor A × B	N	Mean	Std. Dev.
control × f	10	31.14	8.22
drug_a × f	10	34.69	7.22
drug_b × f	10	33.66	8.27
control × m	10	33.81	6.79
drug_a × m	10	33.00	6.86
drug_b × m	10	41.96	4.65

Table 2. ANOVA (main effects + interaction)

Source	SS	df	MS	F	p	partial η^2 [95% CI]
group	307.07	2	153.53	3.04	.056	0.10 [0.00, 1.00]
gender	143.53	1	143.53	2.84	.098	0.05 [0.00, 1.00]
group × gender	250.84	2	125.42	2.48	.093	0.08 [0.00, 1.00]

assumption checks: Levene's & normality of residuals; post-hoc / simple-effects table when an effect is significant. (Levene F=0.51, p=.766 · Shapiro W=0.97, p=.184) — equal variances look reasonable — residual normality looks reasonable

Figure. Interaction



Understanding the numbers

Source. Names which effect this ANOVA row tests - a main effect for one factor, or the A×B interaction between them. Here, the headline effect discussed above is group × gender.

SS. The portion of total variability in the outcome attributable to that row's effect. Here, group × gender has SS = 250.84.

df. The degrees of freedom for that effect - roughly, the number of independent levels being compared. Here, group × gender has df = 2.

MS. The effect's SS divided by its df - the average variability per degree of freedom. Here, group × gender has MS = 125.42.

F. The ratio of an effect's mean square to the residual (error) mean square - larger means the effect explains more variance than chance predicts. Here, F for group × gender = 2.48.

p. The probability of an effect this large (or larger) if that factor/interaction truly had no influence. Here, p = .093 for group × gender - at or above the α = 0.05 threshold.

partial η^2. The share of variance that effect explains after removing variance due to the other effects (~.01 small, .06 medium, .14 large). Here, partial η^2 for group × gender = .08 [.00, 1.00].

Cell (Factor A × B). Each row of the cell-descriptives table names one specific combination of factor levels. Here, the cell descriptives table above breaks the data into 6 such factor-level combinations.

N. The number of observations in that cell. Here, each of the 6 cells above has its own N (see the cell descriptives table for each count).

Mean. The arithmetic average of the outcome within that cell. Here, each cell's mean is listed separately in the cell descriptives table above.

Std. Dev.. How spread out the outcome scores are within that cell. Here, each cell's SD is listed separately in the cell descriptives table above.

M diff. The difference between two cell/level means being compared in the post-hoc / simple-effects table. Here, each row of the simple-effects table above reports its own mean difference for a specific pair.

SE. The standard error of that mean difference. Here, each contrast's SE is listed alongside its mean difference in the simple-effects table above.

p (adj). The pairwise p-value, Tukey-adjusted for the number of comparisons made. Here, each pairwise contrast above carries its own Tukey-adjusted p.

95% CI. The range likely to contain the true mean difference for that pair. Here, each contrast's 95% CI is shown in the simple-effects table above.

How to read this test

Each factor's F/p is its main effect; the A×B row tests whether the effect of one factor depends on the other. A significant interaction usually takes priority — read it from the interaction plot before the main effects. A significant effect for any factor with 3+ levels tells you the levels differ somewhere, not which ones — use the post-hoc / simple-effects (emmeans) table to find the specific pairs. Your significance threshold (α) is 0.05.

APA template: A two-way ANOVA gave A×B interaction $F(2,54)=2.48$, $p = .093$, partial $\eta^2=.08$ [.00, 1.00].

Statistical basis: Factorial ANOVA (main effects + interaction) (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.); Estimated marginal means for interaction / simple-effects follow-up (Lenth RV (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package.); η^2 effect-size benchmarks (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.). Full references in the exported CITATIONS.txt.